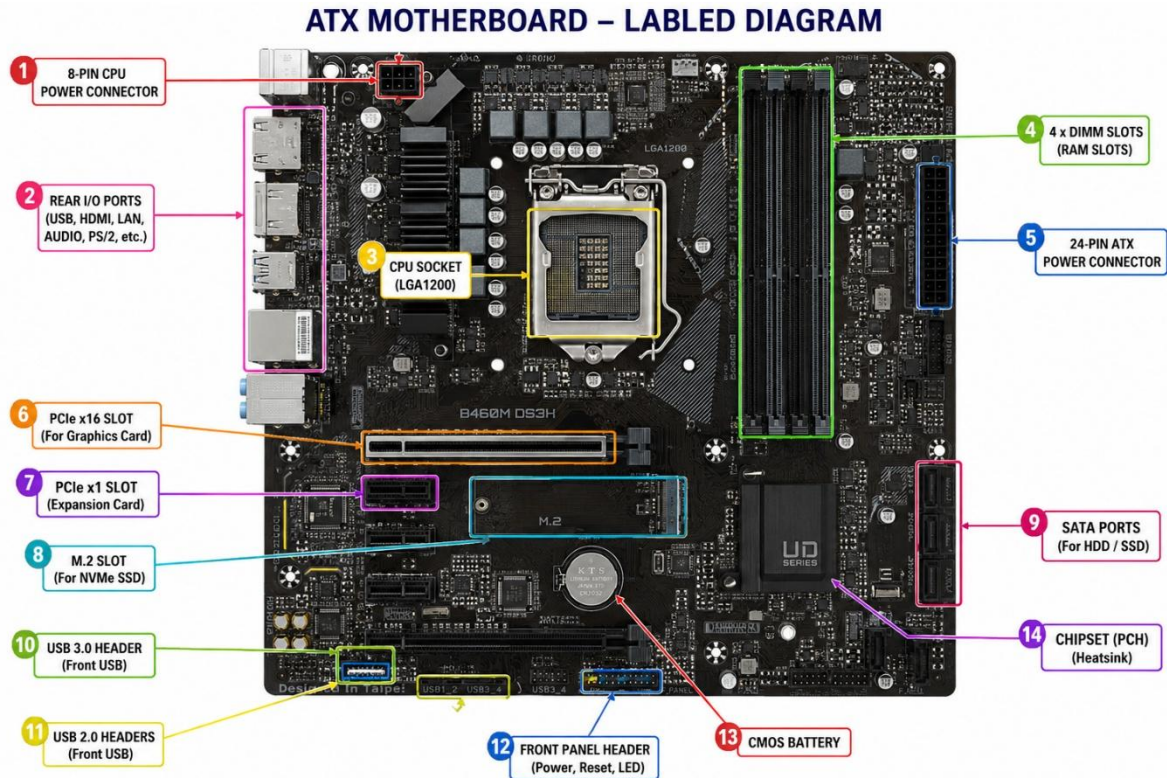


1.

Draw a labeled diagram of an ATX motherboard and mark at least 8 different slots and ports (such as RAM slots, PCIe slots, SATA ports, USB headers, CPU socket, power connector, etc.).

ANS:-



2.

Research and list 3 popular CPU socket types (for example: LGA1151, AM4, LGA1200) and name one compatible CPU model for each.

ANS:-

CPU Socket	Compatible CPU Example	Manufacturer
LGA1151	Intel Core i7-9700K	Intel
AM4	AMD Ryzen 5 5600X	AMD
LGA1200	Intel Core i5-10400	Intel

3.

Explain in your own words the primary functions of the CPU and the chipset on a motherboard, and describe how they work together when you open a music app like Spotify.

ANS:-

CPU (Central Processing Unit)

The CPU is often called the "brain" of the system. It executes instructions, performs calculations, processes data, and controls most operations of the computer. Every application relies on the CPU to carry out its tasks.

Chipset

The chipset is a group of controller chips on the motherboard that manages communication between the CPU and other hardware components such as RAM, storage drives, USB ports, PCIe devices, and networking hardware. It ensures that data flows smoothly between these components.

How They Work Together When Opening Spotify

1. You click the Spotify icon.
2. The CPU receives the command from the operating system.
3. The chipset helps the CPU communicate with the SSD to load Spotify's program files.
4. The chipset also allows data to move between the SSD and RAM.
5. The CPU executes Spotify's instructions from RAM.
6. The chipset manages communication with the sound hardware and network adapter.
7. Music is streamed from the internet, decoded by the CPU, and sent through the audio hardware so you can hear it.

4.

Find a clear photo of a real motherboard (from the internet or your own device), and identify at least 5 different ports or slots by circling them and labeling each one.

ANS:-

